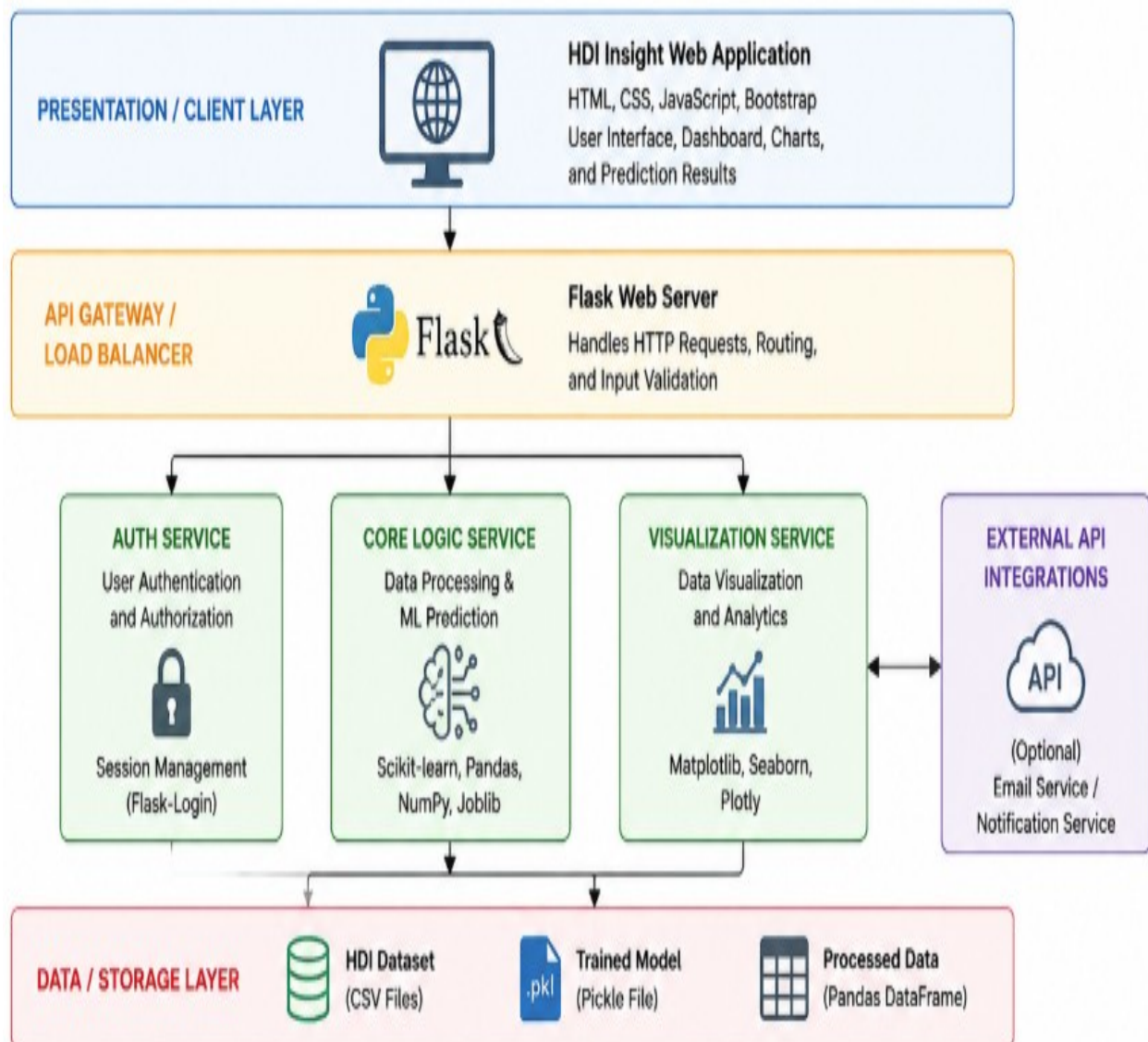


Solution architecture

Date	03 JULY 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum marks	5 Marks

Solution architecture diagram:



Component description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation layer	Provides an interactive user interface where users enter HDI indicators, view dashboards, graphs, and prediction results.	HTML5, CSS3, JavaScript, Bootstrap
API layer (flask server)	Receives user requests, validates input data, communicates with the machine learning model, and returns prediction results.	Python, Flask
Core Logic Service	Performs data preprocessing, loads the trained machine learning model, predicts the HDI category, and prepares visualization data.	Scikit-learn, Pandas, NumPy, Joblib
Visualization Model	Generates charts, graphs, and analytical insights for better understanding of development indicators.	Matplotlib, Seaborn
Data/ Storage Layer	Stores the HDI dataset and trained machine learning model used for prediction and analysis.	CSV Dataset, Pandas DataFrames, Pickle/Joblib